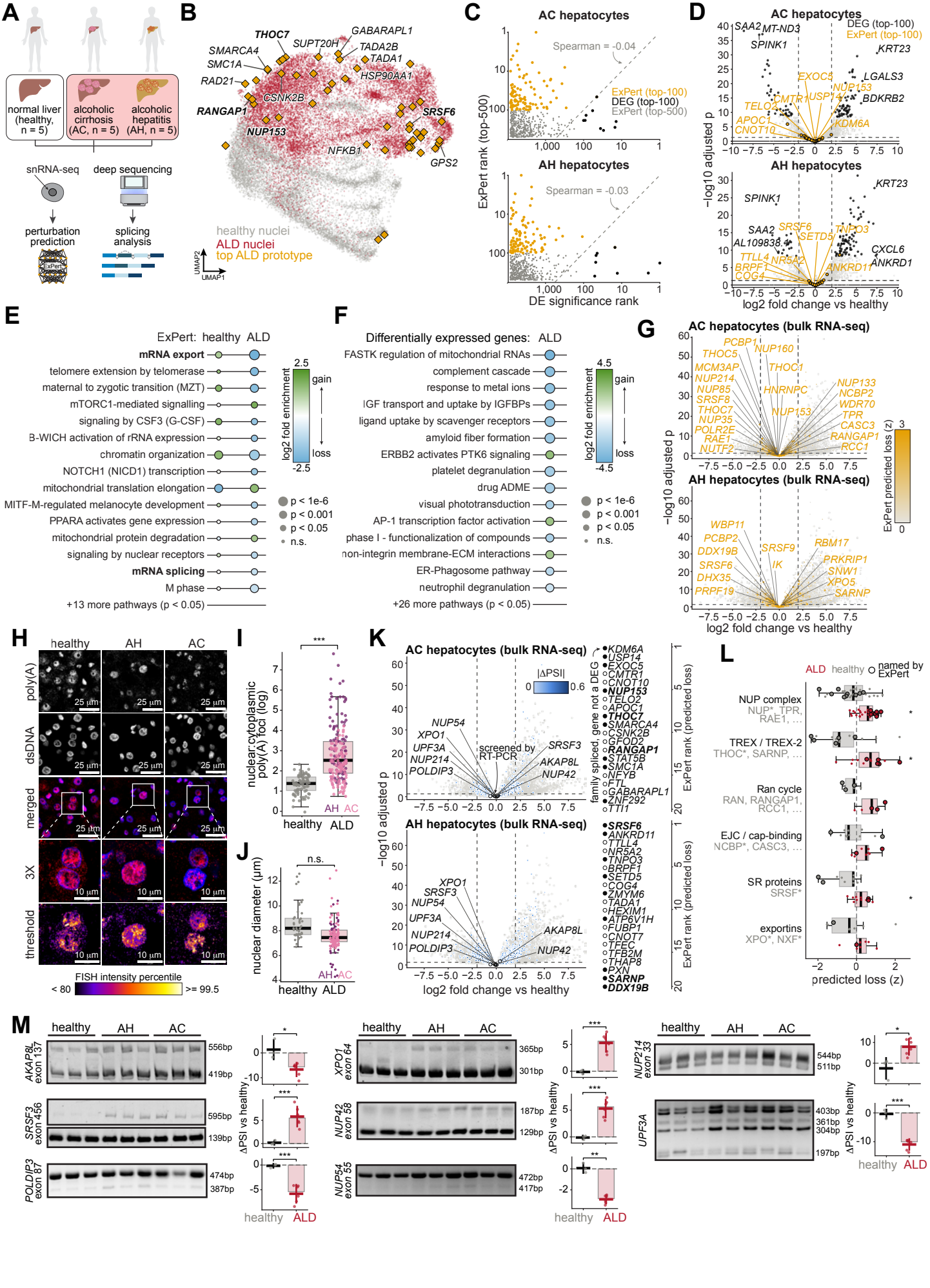
Figure 5


A
normal liver (healthy, n = 5)
alcoholic cirrhosis (AC, n = 5)
alcoholic hepatitis (AH, n = 5)
snRNA-seq
deep sequencing
perturbation prediction
splicing analysis
B
THOC7
SUPT20H
GABARAPL1
SMARCA4
TADA2B
SMC1A
TADA1
RAD21
HSP90AA1
CSNK2B
RANGAP1
SRSF6
NUP153
NFKB1
GPS2
healthy nuclei
ALD nuclei
top ALD prototype
UMAP2
UMAP1
C
AC hepatocytes
Spearman = -0.04
ExPert (top-100)
DEG (top-100)
ExPert (top-500)
ExPert rank (top-500)
AH hepatocytes
Spearman = -0.03
DE significance rank
D
AC hepatocytes
DEG (top-100)
ExPert (top-100)
SAA2
MT-ND3
SPINK1
KRT23
EXOC5
LGALS3
NUP153
USP14
CMTR1
BDKRB2
TELO2
APOC1
CNOT10
KDM6A
AH hepatocytes
SRSF6
SETD5
TNPO3
AL109838.4
CXCL6
ANKRD1
TTLL4
NR5A2
BRPF1
COG4
−log10 adjusted p
log2 fold change vs healthy
E
ExPert: healthy ALD
mRNA export
telomere extension by telomerase
maternal to zygotic transition (MZT)
mTORC1-mediated signalling
signaling by CSF3 (G-CSF)
B-WICH activation of rRNA expression
chromatin organization
NOTCH1 (NICD1) transcription
mitochondrial translation elongation
MITF-M-regulated melanocyte development
PPARA activates gene expression
mitochondrial protein degradation
signaling by nuclear receptors
mRNA splicing
M phase
+13 more pathways (p < 0.05)
log2 fold enrichment
gain
loss
p < 1e-6
p < 0.001
p < 0.05
n.s.
F
Differentially expressed genes: ALD
FASTK regulation of mitochondrial RNAs
complement cascade
response to metal ions
IGF transport and uptake by IGFBPs
ligand uptake by scavenger receptors
amyloid fiber formation
ERBB2 activates PTK6 signaling
platelet degranulation
drug ADME
visual phototransduction
AP-1 transcription factor activation
phase I - functionalization of compounds
non-integrin membrane-ECM interactions
ER-Phagosome pathway
neutrophil degranulation
+26 more pathways (p < 0.05)
G
AC hepatocytes (bulk RNA-seq)
PCBP1
NUP160
THOC5
MCM3AP
THOC1
NUP214
NUP85
HNRNPC
NUP133
NCBP2
SRSF8
WDR70
THOC7
NUP153
TPR
NUP35
CASC3
POLR2E
RANGAP1
RAE1
RCC1
NUTF2
AH hepatocytes (bulk RNA-seq)
WBP11
PCBP2
SRSF9
RBM17
DDX19B
PRKRIP1
IK
SRSF6
SNW1
DHX35
XPO5
PRPF19
SARNP
ExPert predicted loss (z)
H
healthy
AH
AC
poly(A)
dsDNA
merged
3X
threshold
25 µm
10 µm
FISH intensity percentile
< 80
>= 99.5
I
nuclear:cytoplasmic poly(A) foci (log)
***
healthy
ALD
J
nuclear diameter (µm)
n.s.
K
AC hepatocytes (bulk RNA-seq)
|ΔPSI|
family spliced, gene not a DEG
KDM6A
USP14
EXOC5
CMTR1
CNOT10
NUP153
TELO2
APOC1
THOC7
SMARCA4
CSNK2B
GFOD2
RANGAP1
STAT5B
SMC1A
NFYB
FTL
GABARAPL1
ZNF292
TTI1
NUP54
XPO1
UPF3A
NUP214
POLDIP3
screened by RT-PCR
SRSF3
AKAP8L
NUP42
ExPert rank (predicted loss)
AH hepatocytes (bulk RNA-seq)
SRSF6
ANKRD11
TTLL4
NR5A2
TNPO3
BRPF1
SETD5
COG4
ZMYM6
TADA1
HEXIM1
ATP6V1H
FUBP1
CNOT7
TFEC
TFB2M
THAP8
PXN
SARNP
DDX19B
log2 fold change vs healthy
L
ALD
healthy
named by ExPert
NUP complex
NUP*, TPR, RAE1, …
TREX / TREX-2
THOC*, SARNP, …
Ran cycle
RAN, RANGAP1, RCC1, …
EJC / cap-binding
NCBP*, CASC3, …
SR proteins
SRSF*
exportins
XPO*, NXF*
predicted loss (z)
M
healthy
AH
AC
AKAP8L exon 137
556bp
419bp
SRSF3 exon 456
595bp
139bp
POLDIP3 exon 87
474bp
387bp
ΔPSI vs healthy
ALD
XPO1 exon 64
365bp
301bp
NUP42 exon 58
187bp
129bp
NUP54 exon 55
472bp
417bp
NUP214 exon 33
544bp
511bp
UPF3A
403bp
361bp
304bp
197bp